

Lecture 3: Pretrained Networks

Overview

- Pretrained networks
 - Image recognition – a pretrained network that recognizes the subject of an image
 - Image-to-image translation – a pretrained model that fakes it until it makes it

Pretrained Networks

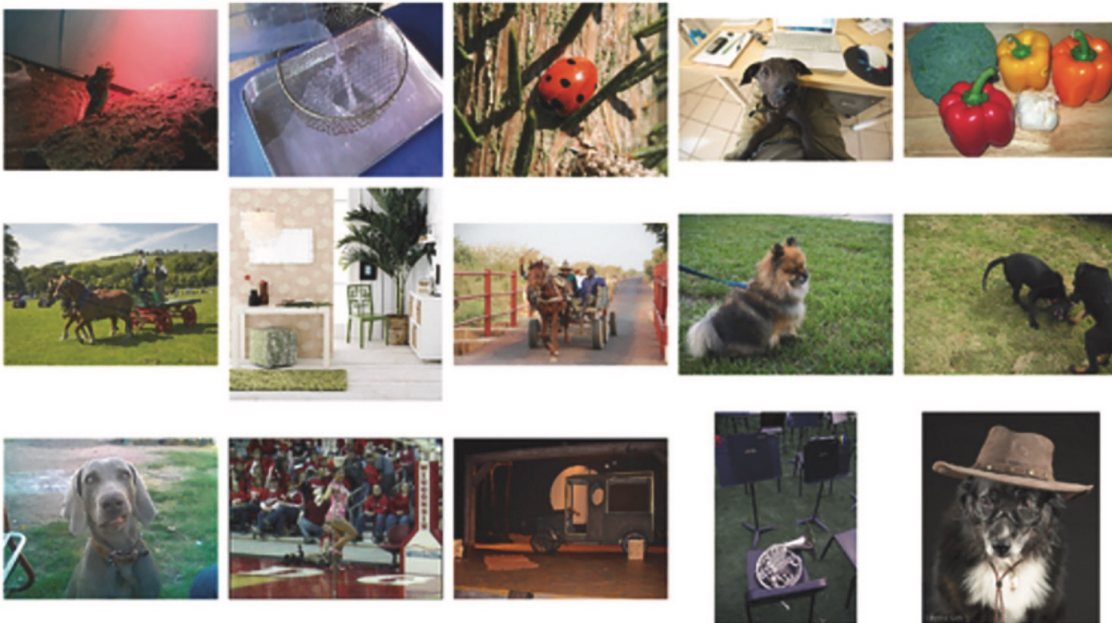
- Pretrained Network refer to a neural network that has been previously trained on a dataset and learned to extract powerful and informative features.
- It is common for researchers to publish their source code along with their papers, and often the code comes with weights that were obtained by training a model on a reference dataset.
- Using one of these models could enable us to, for example, equip our next web service with image-recognition capabilities with very little effort.

Pretrained Networks

- Benefits of pretrained networks:
 - Easy to implement
 - Good performance (as a starting point)
 - Labeled data is not required
 - No model training process
- Disadvantages:
 - Not suitable for a different data set or task
 - Different data preprocessing
 - Require transfer and fine tune

Task 1: Image Recognition

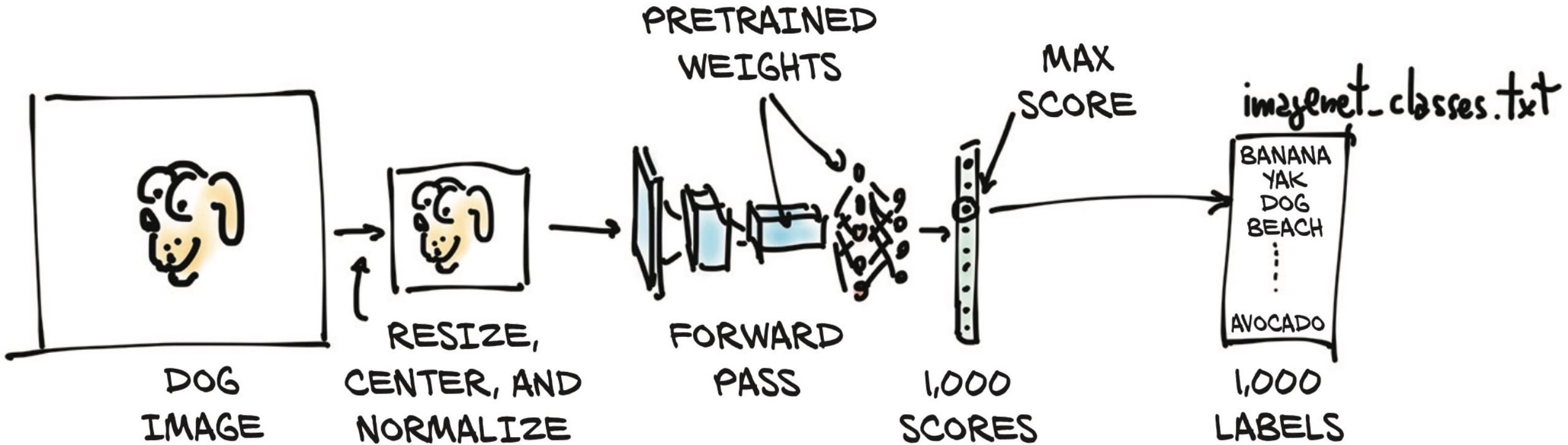
- The image-classification task: given an input image and producing an image label out of 1,000 total categories, describing the content of the image.



The training set for ImageNet Large Scale Visual Recognition Challenge (ILSVRC) consists of 1.2 million images labeled with one of 1,000 nouns (for example, “dog”), referred to as the *class (label)* of the image.

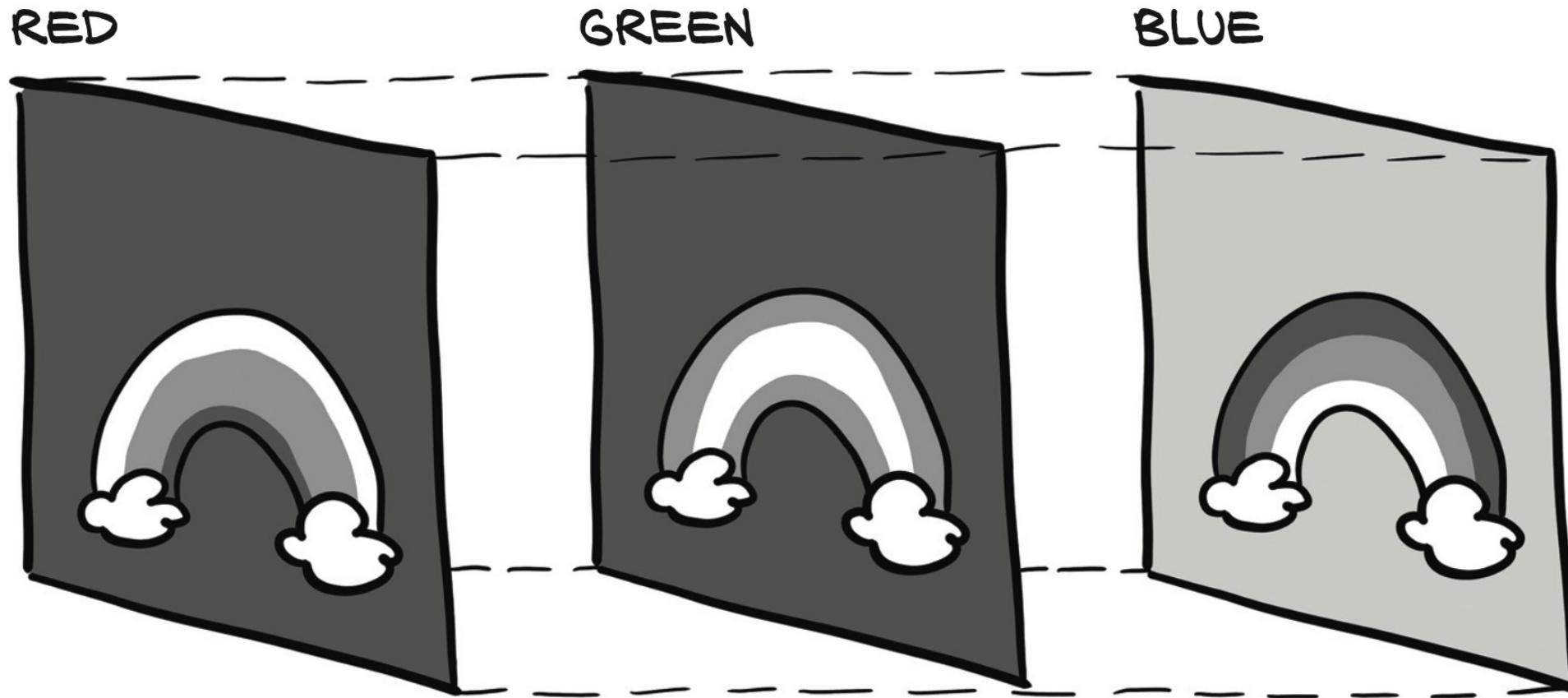
A small sample of ImageNet images

Task 1: Image Recognition



- Image -> tensors -> image-classes

Task 1: Image Recognition



RGB image is represented as a three-dimensional tensor: height, width, and the three-color channels

Each of the RGB channels captures a certain portion of the spectrum

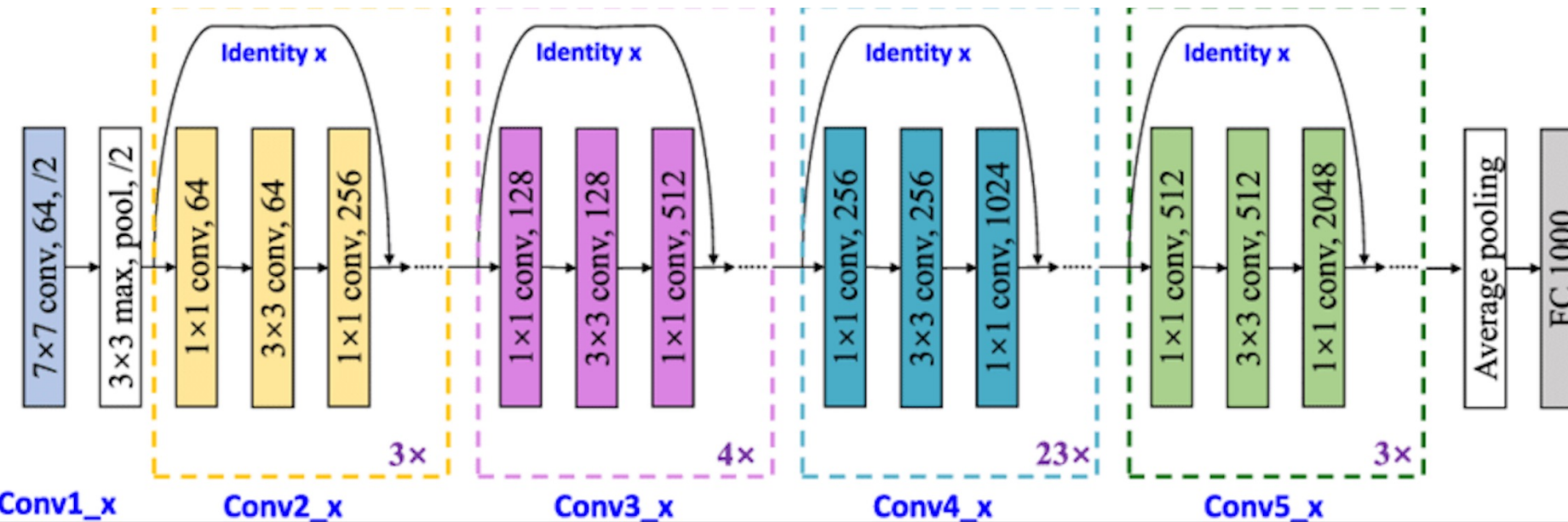
Task 1: Image Recognition

- Our model takes a processed image and pass it into the pretrained network to obtain scores for each class
 - The highest score corresponds to the most likely class according to the weights
- This will result in a list of predicted labels for that image, which we can then examine to see what the model thinks our image is.

Task 1: Image Recognition – Resnet

- The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) has gained popularity since its inception in 2010.
- An ensemble of residual nets achieves 3.57% error on the ImageNet test set. This result won the 1st place on the ILSVRC 2015 classification task.

Task 1: Image Recognition – Resnet



Pretrained resnet101: the weights of resnet101 trained on the ImageNet dataset, with 1.2 million images and 1,000 categories.

Resnet101 sports 44.5 million parameters

From: https://www.researchgate.net/figure/The-structure-of-the-ResNet-101-based-deep-feature-extractor_fig4_348078198

Task 1: Image Recognition – Resnet

- Step 1: loading an image



Task 1: Image Recognition – Resnet

Step 2

- Preprocess the images: scale the input image to 256×256 , crop the image to 224×224 around the center, transform it to a tensor, and normalize its RGB (red, green, blue) components to match what was presented to the network during training

Task 1: Image Recognition – Resnet

- Producing a vector of 1,000 scores, one per ImageNet class

```
tensor([[ -3.4803,  -1.6618,  -2.4515,  -3.2662,  -3.2466,  -1.3611,
          -2.0465,  -2.5112,  -1.3043,  -2.8900,  -1.6862,  -1.3055,
          ...,
          2.8674,  -3.7442,   1.5085,  -3.2500,  -2.4894,  -0.3354,
          0.1286,  -1.1355,   3.3969,   4.4584]])
```

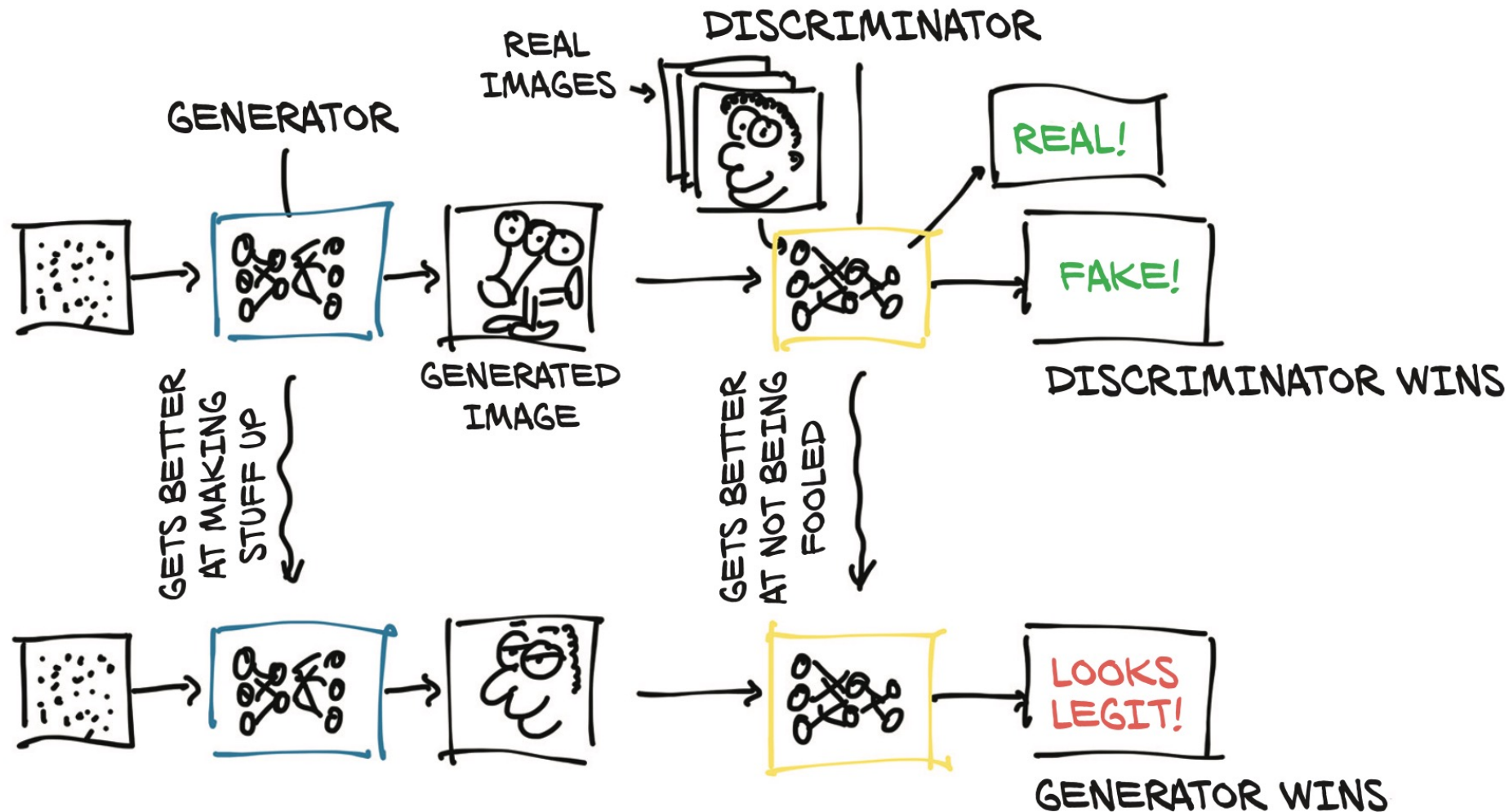
Task 1: Image Recognition – Resnet

- Find out the label of the class that received the highest score.
 - (tensor([207]), 'golden retriever', 96.29335021972656)
- Sorts the values
 - [('golden retriever', 96.29335021972656),
 - ('Labrador retriever', 2.8081188201904297),
 - ('cocker spaniel, English cocker spaniel, cocker', 0.2826734781265259),
 - ('redbone', 0.2086300402879715),
 - ('tennis ball', 0.11621593683958054)]
- This is a great example of the fundamental differences in how humans and neural networks view the world

Task 2: Image-to-Image Translation—GAN

- GAN stands for *generative adversarial network*, where *generative* means something is being created (in this case, fake masterpieces), *adversarial* means the two networks are competing to outsmart the other.
- The *generator* network takes the role of the painter, tasked with producing realistic-looking images, starting from an arbitrary input. The *discriminator* network is the amoral art inspector, needing to tell whether a given image was fabricated by the generator or belongs in a set of real images. This two-network design is atypical for most deep learning architectures but, when used to implement a GAN game, can lead to incredible results.

Task 2: Image-to-Image Translation—GAN

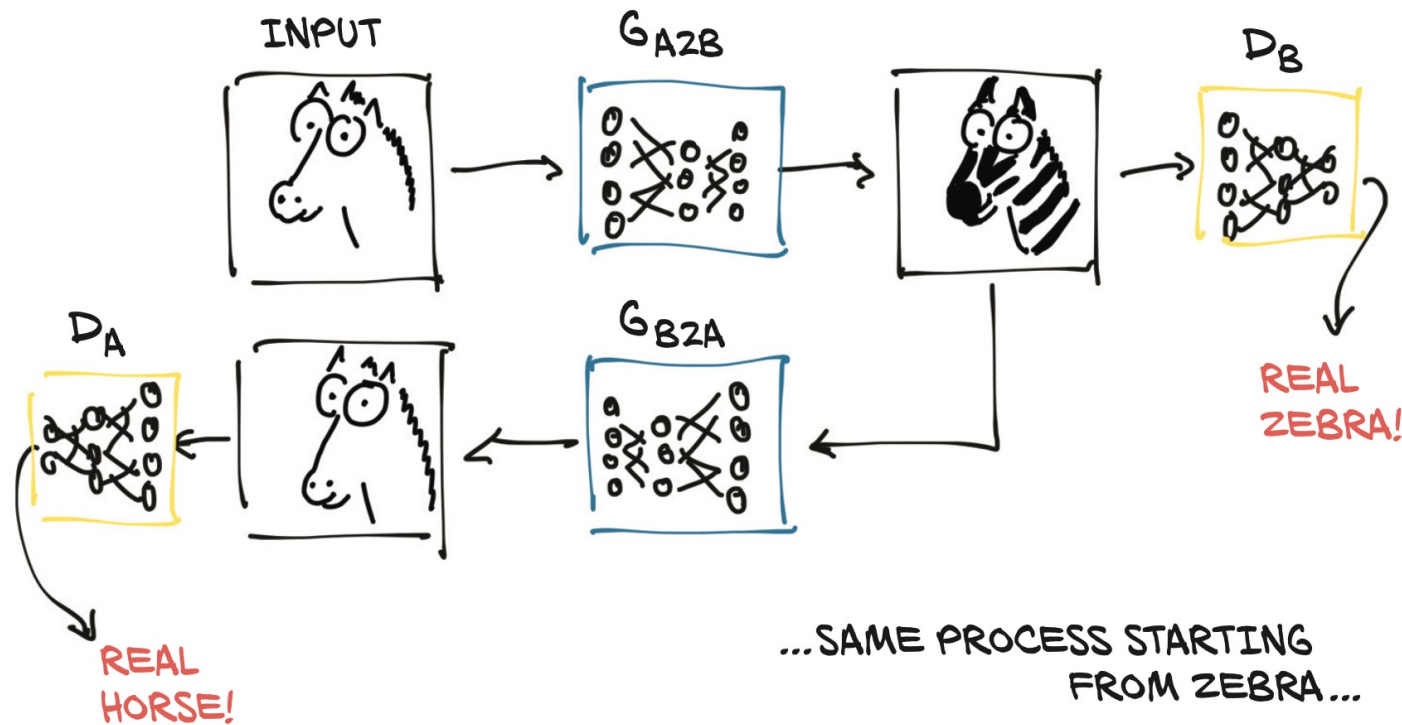


Task 2: Image-to-Image Translation—GAN

- The end goal for the generator is to fool the discriminator into mixing up real and fake images.
- The end goal for the discriminator is to find out when it's being tricked, but it also helps inform the generator about the identifiable mistakes in the generated images
- A well-trained generator learns a plausible model for generating images that look real even when examined by humans.

CycleGAN

- A CycleGAN can turn images of one domain into images of another domain (and back)
- CycleGAN workflow for the task of turning a photo of a horse into a zebra, and vice versa



Task 2: Image-to-Image Translation—GAN



Task 2: Image-to-Image Translation—GAN

- Generative models are also being explored for producing real-sounding audio, credible text, and enjoyable music. It is likely that these models will be the basis of future tools that support the creative process.

AI-generated Images

- In fact, there's even a market for AI's original artwork—Google hosted an art show to benefit charity and to showcase work created by its software DeepDream. It sold an AI-generated piece of art that was a collaboration between human and machine for \$8,000 plus others.



British auction house Christie sold its first piece of AI-generated art, “Portrait of Edmond Belamy,” for \$432,500.



Generative Models

- Deep Learning
 - Generative adversarial network (GAN) — a class of neural network — to generate some extremely realistic faces.



Generative Models



Summary

- A pretrained network is a model that has already been trained on a dataset. Such networks can typically produce useful results immediately after loading the network parameters.
- By knowing how to use a pretrained model, we can integrate a neural network into a project without having to design or train it.
- ResNet: a deep convolutional neural network.
- Generative adversarial networks (GANs) have two parts—the generator and the discriminator—that work together to produce output indistinguishable from authentic items.
- CycleGAN uses an architecture that supports converting back and forth between two different classes of images.